

Ivan Ball-llovera

SENIOR SOFTWARE ARCHITECT

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PROFESSIONAL SUMMARY

Senior Software Architect with 25+ years designing and delivering scalable, cloud-native systems on the Microsoft stack. Deep expertise in Domain-Driven Design, Clean Architecture, CQRS, and microservices, paired with hands-on Azure cloud architecture and DevOps (Bicep, Docker, Kubernetes, CI/CD). Proven record of leading architectural refactors, modernizing legacy platforms, and mentoring cross-functional teams to raise engineering standards. Currently architecting production-grade .NET 10 platforms spanning modular-monolith-to-microservices, event-driven messaging, and AI/RAG integration, while leveraging AI-assisted development workflows to ship faster without sacrificing quality.

CORE COMPETENCIES

Architecture & Design Domain-Driven Design, Clean Architecture, CQRS, Modular Monolith → Microservices, Event-Driven Architecture, Outbox Pattern, gRPC, API Gateway (YARP), Design Patterns

Cloud & DevOps Azure (Container Apps, Service Bus, Cosmos DB, SQL, App Insights, Log Analytics), Infrastructure as Code (Bicep, ARM), Docker, Kubernetes, Service Fabric, CI/CD (GitHub Actions, Azure DevOps), OIDC / Managed Identity

Languages & Frameworks .NET (Core through 10) / C#, Blazor, .NET MAUI, ASP.NET Core Web API, .NET Aspire, EF Core, Angular, TypeScript, Python

Data & BI SQL Server, Cosmos DB, MySQL, Oracle, Data Engineering, Data Warehousing, SSIS, SSAS (Tabular), SSRS, Power BI

Quality & Security xUnit, NUnit, Moq, Playwright, Selenium, TDD, static analysis & application security (SAST/DAST, Snyk, Fortify, Wiz)

Leadership & Practices Technical leadership, mentorship, Agile / Scrum / Kanban, architecture governance (ADRs), cross-team collaboration, AI-assisted development (Claude Code, GitHub Copilot)

FEATURED PROJECT · MMCA PLATFORM

Personal / Open Source · github.com/ivanball/MMCA.Common

A production-grade .NET 10 reference platform demonstrating modern enterprise architecture end-to-end. The conference application was deployed to Azure and powered a live Atlanta Developers Conference.

- Designed a modular monolith that extracts cleanly into independent microservices: modules implement a discovery contract, load in topological order, and run behind a YARP gateway with gRPC contracts and typed clients, keeping transport reversible at the edge.
- Authored a 12-package shared framework (DDD, Clean Architecture, CQRS) published to GitHub Packages and consumed by an e-commerce app (Stripe) and a conference app.
- Implemented database-per-service with an at-least-once outbox plus dual dispatch, soft-delete, automatic audit stamping, and GDPR/CCPA anonymization; documented decisions with 9 ADRs and a 34-category architecture-review rubric.
- Deployed to Azure using Bicep IaC, Service Bus, Container Apps, App Insights, and Log Analytics, with OIDC / Managed-Identity authentication; orchestrated locally with .NET Aspire.
- Built Blazor (MudBlazor) and .NET MAUI clients; enforced quality with xUnit v3, Playwright E2E, and automated accessibility testing (axe, WCAG 2.1 AA: zero violations).

PROFESSIONAL EXPERIENCE

Senior Software Engineer • Assurant

June 2025 – Present

Architect-level scope: platform, security, and cross-team technical decisions

- Drove architecture and platform decisions on Assurant's Renters Insurance platform (Angular, C#, Azure Service Fabric microservices, SQL Server), delivering multi-factor authentication (MFA) across the FlexDeposit and SureDeposit products and serving as a trusted technical authority across the Renters organization.
- Established testing and quality strategy with QA toward an 80% new-code coverage target, and set application-security standards, remediating Fortify SAST/DAST findings and instituting stricter code-review gates to sustain a zero-new-vulnerability baseline.
- Architected "A.I.M." (AI Initiative & Modernization) at a company hackathon, an AI assistant embedded in Azure DevOps spanning the full delivery lifecycle (plan → build → test → ship → operate): 20 capabilities across 6 squad roles with a multi-provider, human-in-the-loop design ("AI suggests, humans decide") and integrations across GitHub, Application Insights, Datadog, and Confluence.
- Guided architectural decisions and engineering best practices and mentored engineers; authored key system documentation in Confluence to improve maintainability and shared understanding.

Principal Software Engineer • Hitachi Solutions America

May 2021 – May 2025

- Led the architectural refactor of biBERK's cloud-native insurance platform (C#, Angular, Azure SQL, Cosmos DB), modernizing the system and establishing scalable patterns adopted across the organization.
- Served as the go-to architect for complex design challenges across biBERK, Empower, Analog Devices, and Cost Management engagements; coached engineers on methods, tooling, and best practices.
- Authored cross-cutting design specifications (e.g., the Empower Connections page layout engine spanning UI, API, and database) and built CI/CD pipelines using Infrastructure as Code (ARM templates).
- Developed a document-ingestion module in Python for a Retrieval-Augmented Generation (RAG) architecture powering internal AI initiatives.

Technical Lead & Senior Software Engineer • Ultimate Software

December 2015 – May 2021

- Architected a PostUpgrade automated testing engine (.NET Core, Angular, Selenium) and defined the architecture for the CAT application suite, heading off ~32 customer escalations and saving an estimated 80 hours of support time.
- Led the migration of 11 production database shards from MariaDB (Pivotal Cloud Foundry) to MySQL on Kubernetes, adding Slack-integrated yellow/red monitoring and alerting.
- Re-engineered WFM ETL processing (SSIS) across all data centers, cutting execution time from 2+ hours to under an hour; established engineering best practices within Agile/Scrum.

Software Architect & Lead Developer • Miami-Dade County

August 2009 – December 2015

Property Appraiser's Office

- Built a reusable .NET application framework enabling a team to deliver line-of-business applications (ASP.NET Web Forms N-Tier, then ASP.NET MVC with Visual Studio 2013, HTML5/CSS3/JavaScript/Bootstrap, and Entity Framework 6).
- Architected the SQL Server backend for the County's most-trafficked web app (Property Search, 250,000+ hits/day), using advanced partitioning across multi-cluster, load-balanced servers for parallel query optimization.
- Implemented an enterprise Data Warehouse loading SQL Server from multiple Oracle sources (re-architected from a multithreaded C# loader to SSIS), surfacing analytics through an SSAS Tabular model and SSRS reports; administered 20+ in-house SQL Server databases.

COMMUNITY LEADERSHIP

One of the lead organizers for two community-driven Atlanta technology conferences:

- **Atlanta Cloud + AI Conference**
- **Atlanta Developers Conference**

Coordinated speakers, sponsors, volunteers, event logistics, marketing, and attendee experience to deliver successful conferences focused on cloud computing, artificial intelligence, and software engineering. Both conferences ran on the MMCA conference application (see Featured Project).

EDUCATION

B.S., Computer Science

University of Havana (Faculty of Mathematics)
Havana, Cuba · 1994 – 1999

LANGUAGES

English · Spanish (bilingual)

CERTIFICATIONS

- **Microsoft Certified:** Azure Administrator Associate (AZ-104, 2025); Azure AI Fundamentals (AI-900, 2024); Azure Data Fundamentals (DP-900, 2021); Azure Fundamentals (AZ-900, 2021)
- **MCSA · SQL Server 2012:** Querying, Administering, and Implementing a Data Warehouse (70-461 / 70-462 / 70-463, 2013)
- **MCSA: Web Applications (2016):** Developing ASP.NET MVC 4 (70-486), Programming in HTML5/JavaScript/CSS3 (70-480), Programming in C# (70-483)
- **Earlier Microsoft certifications (2006–2009):** .NET Framework 3.5: Foundation, WCF, and Windows Forms (70-536 / 70-503 / 70-505); Solution Architecture (70-300); C# Web, Windows, and XML Web Services (70-315 / 70-316 / 70-320); SQL Server 2000 databases (70-229)